



Energy-Efficient Design and Delay Optimization of a Absolute-Value Detector

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Abstract. With the rapid development of digital integrated circuits, absolute value detectors have become indispensable devices in signal processing, control systems, arithmetic operations and other fields. This paper presents an optimal design for a 4-bit absolute value detector with a focus on reducing latency and power consumption using mixed logic methods and logical effort analysis. The circuit combines static CMOS logic for control and through-tube logic (PTL) for data processing to achieve energy efficiency and high speed. XOR gates, full adders, and multiplexers are used to perform conditional inversion and 2's complement operations based on sign bits. A three-input comparator is integrated into the same logical stream, minimizing redundancy and area. Critical path analysis and gate sizing with logical effort help minimize propagation delays and improve timing performance. Logic simplification through the Carnot map also reduces the complexity of the circuit. The final design proves the balance between speed, precision and energy efficiency, making it ideal for modern VLSI systems. This research provides a practical framework for efficient digital design and opens up possibilities for future enhancement using advanced technologies such as finfet, CNTs or 2D materials.

Keywords: Absolute-Value Detector, Hybrid Logic, Logical Effort, Propagation Delay

1 Introduction

With the rapid development of digital integrated circuits, absolute-value detectors have become increasingly important in applications such as signal processing, control systems, and arithmetic operations. These circuits are critical in enhancing the accuracy, performance, and energy efficiency of embedded systems. As power and area constraints become more stringent in modern VLSI design, there is a growing need for optimized solutions that balance speed, power consumption, and silicon real estate. Prior studies, such as the work by Rabaey et al., emphasize that the integration of low-power design methodologies is key to advancing high-performance digital systems under limited energy budgets [1]. In addition, recent research by Bhunia and Mukhopadhyay highlights the growing significance of logic-level optimization techniques in minimizing delay and leakage power in nanoscale technologies,

underscoring the need for compact and energy-efficient circuit architectures like the one proposed in this paper [2]. The effectiveness of PTL in reducing transistor count and switching activity has also been emphasized by Zimmermann and Fichtner, who compared CMOS with pass-transistor logic in low-power arithmetic circuits [3]. Wu and Liu further validated that PTL-based arithmetic components are advantageous for embedded systems, where energy efficiency and compactness are critical [4]. Similarly, in a 2025 study, Yang proposed an optimization framework that systematically reduced propagation delay in high-speed digital circuits by applying logical effort theory and tailored gate sizing strategies [5]. This work demonstrated that critical-path analysis, when combined with power-aware logic restructuring, could significantly enhance both speed and energy efficiency in CMOS-based systems. Furthermore, Wang's recent design of a CMOS-based 4-bit absolute-value detector illustrated the practical benefits of integrating comparator logic directly into the main data path, leading to reduced gate count and lower dynamic power without compromising accuracy [6]. To support delay estimation and design trade-offs, the logical effort framework plays an essential role in early-stage circuit evaluation. Sutherland, Sproull, and Harris proposed a widely adopted method that uses path logical effort and electrical effort to predict and balance delay across critical paths [7].

This paper investigates methods for improving circuit performance in the context of a 4-bit absolute-value detector, with a focus on reducing delay and power usage through logical and structural optimization. The structure of this paper is organized as follows: Chapter 2 focuses on the logic gate circuit design, including its definition, complete schematic, and logic optimization techniques; Chapter 3 discusses the critical path of the circuit and how it affects performance; Chapter 4 presents the overall optimization strategies applied at various levels, including logic, circuit, and layout. The goal of this study is to provide a comprehensive design framework for developing efficient absolute-value detectors suitable for low-power, high-speed applications.

2 Design Process

2.1 Principle and design requirements

A 4-bit absolute value detector is a digital circuit used to convert a 4-bit signed binary number to the corresponding absolute value (non-negative). In the complement representation of 2, the most significant bit (MSB) represents the symbol of the number – “0” means non-negative and “1” means negative. For example, the 4-bit binary number 1001 represents -7, while 0111 represents +7 [8]. The goal of an absolute value detector is to output an unsigned amplitude regardless of the input symbol. Essentially, if the input number is positive or zero, it is passed unchanged; If the number is negative, the output becomes its 2's complement.

The complete circuit diagram of the 4-bit absolute value detector integrates symbol detection, conditional transformation, and output logic into an efficient and accurate cohesive system. The architecture consists of two main functional modules: a main four-input circuit and a supported three-input comparator. A four-input circuit handles four bits of signed input (A_3, A_2, A_1, A_0), where A_3 acts as a sign bit and determines whether the number is positive or negative. If A_3 is high, indicating a negative number,

the circuit conditionally activates the conversion process by reversing the bits and adding 1 according to the 2-complement method of A2, A1, and A0 [9]. The logical design must determine whether the input is negative, and if it is, generate a complement of 2. This process involves inverting the lower three digits (A2, A1, A0) and adding “1” to the result.

The logic gates should be implemented to minimize the number of transistors and switching delay. To achieve this, the design takes a hybrid approach, using static CMOS logic for the control path and using the transistor logic (PTL) for efficient computational bit operations. In addition, logical expressions are optimized using Carnot maps (K-Maps) to reduce gate complexity and power consumption. The truth table 1 is shown below.

Table 1. Truth table

A	A	A	A	Decima	F	F	F	Decima
3	2	1	0	1	2	1	0	1
0	0	0	0	+0	0	0	0	0
0	0	0	1	+1	0	0	1	1
0	0	1	0	+2	0	1	0	2
0	0	1	1	+3	0	1	1	3
0	1	0	0	+4	1	0	0	4
0	1	0	1	+5	1	0	1	5
0	1	1	0	+6	1	1	0	6
0	1	1	1	+7	1	1	1	7
1	0	0	0	-8	0	0	0	0
1	0	0	1	-7	0	0	1	7
1	0	1	0	-6	0	1	0	6
1	0	1	1	-5	0	1	1	5
1	1	0	0	-4	1	0	0	4
1	1	0	1	-3	1	0	1	3
1	1	1	0	-2	1	1	0	2
1	1	1	1	-1	1	1	1	1

The circuit design Is basically in “orme” by the truth table, which Outlines the mapping of all 16 possible 4-bit signed inputs to the corresponding absolute values. For example, 1001 (−7) maps to 0111(+7), while 0000 remains unchanged. This mapping ensures that the output is always non-negative, thus verifying the correctness of the circuit function. Karnaugh Maps (K-Maps) are then used to simplify the Boolean expressions derived from this truth table, reducing logical complexity and increasing efficiency. The logical operation formulas are as follows.

$$\begin{aligned}
 F_0 &= A_0 \\
 F_1 &= \overline{A_3}A_1 + A_3(A_1 \oplus A_0) \\
 F_2 &= \overline{A_3}A_2 + A_3(\overline{A_2} \oplus (A_1))
 \end{aligned}
 \tag{1}$$

2.2 Logic gate design

To achieve this transformation, the design uses XOR gates for conditional inversion, each controlled by a NegFlag signal from A3. These outputs are then fed into a series of three complete adders, which implement the controlled addition of binary "1" and complete the absolute value calculation of the negative input. In contrast, for a non-negative input ($A_3 = 0$), the original bit passes directly without modification. In the final stage, a multiplexer (MUX) is used to select between the converted signal and the unaltered signal based on the value of the sign bit, ensuring the correct absolute value of the output. Logic gate is shown in Figure 1.

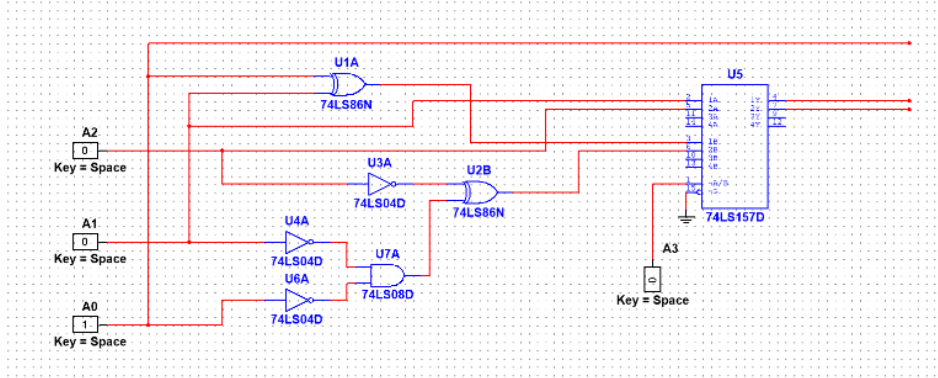


Fig. 1. Logic gate (Photo/Picture credit: Original)

Bit Absolute Value Converter. In addition to the main four-input circuit, the system includes a three-input comparator that evaluates whether the amplitude portion of the input (A_2, A_1, A_0) is greater than another 3-bit input (usually zero). This comparator plays a role in supporting internal control logic and can be reused in extended applications, such as signed magnitude comparison or overflow detection. The comparator also uses an optimized logic gate design, based on its own truth table and Boolean simplification process.

$$A > B = A_2 \overline{B_2} + \left[(\overline{A_2} B_2) + (A_2 B_2) \right] \left[A_1 \overline{B_1} + \left[(\overline{A_1} B_1) + (A_1 B_1) \right] \left[A_0 \overline{B_0} \right] \right] \quad (2)$$

Boolean expression. In the design of digital circuits, Boolean expressions play a crucial role in defining the logical operations that control how the input is converted to the output [10]. In this section, we will focus on discussing the Boolean expressions derived from the truth table of the 4-bit absolute value detector, which helps to minimize the complexity of logical design. By simplifying the Boolean expression, we can reduce the number of required logic gates, thereby improving the overall circuit efficiency in terms of speed and power consumption.

The truth table 2 given below Outlines the relationship between the inputs (A_2, A_1, A_0) and the corresponding outputs, explaining how each combination of inputs determines the logic required for inversion and addition operations. These expressions are crucial for further optimizing the circuit design using the k-map, enabling us to find the simplest logic to achieve the required functions.

Table 2. Truth table

A	B							
	000	001	010	011	100	101	110	111
000	0	0	0	0	0	0	0	0
001	1	0	0	0	0	0	0	0
010	1	1	0	0	0	0	0	0
011	1	1	1	0	0	0	0	0
100	1	1	1	1	0	0	0	0
101	1	1	1	1	1	0	0	0
110	1	1	1	1	1	1	0	0
111	1	1	1	1	1	1	1	0

To improve overall performance, the circuit uses a hybrid logic style: due to its high noise resistance and reliable switching behaviour, static CMOS is used for control logic and via transistor logic (PTL) for data paths to reduce transistor count and energy efficiency. The Logical Effort (LE) method is used to further refine the size of the logical gates, which helps balance latency between different paths and minimizes critical path bottlenecks. The resulting circuit, shown in Figure 2, provides a compact, efficient, and scalable solution for absolute value calculation in digital systems.

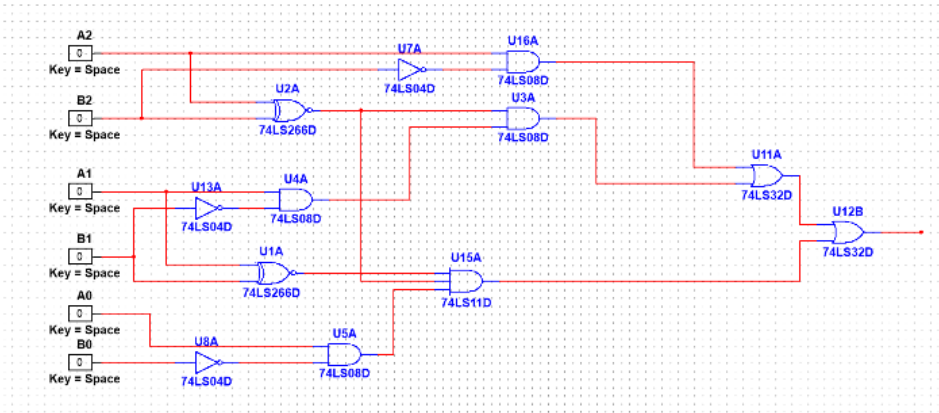


Fig. 2. The resulting circuit (Photo/Picture credit: Original)

2.3 Optimization

In order to improve circuit efficiency and reduce redundancy, the 4-bit absolute value detector and 3-bit comparator are integrated into a unified circuit. The comparator is not implemented as a separate module, but rather has its logic embedded in the main absolute value computation stream. Both circuits rely on shared input bits (A2, A1, A0) and symbol bits (A3), allowing intermediate signals to be reused without the need to duplicate logic gates. For example, the XOR gates used to conditionally invert A2, A1, and A0 in absolute value calculations can also support comparison logic, eliminating the need for a separate invert circuit in the comparator. Similarly, the control signal derived from A3 (which is used to determine whether to apply the complement of 2) is also used to control the comparison behaviour.

By aligning logical paths and identifying overlapping operations, the two circuits are combined into a single architecture that performs both functions simultaneously. This integration simplifies the signal flow and reduces the number of gates, resulting in a more compact layout and lower power consumption. In addition, the combined design improves timing coordination, as both absolute values and comparison results are generated from the same logical sequence without increasing latency. Overall, this optimization improves the performance and area efficiency of the circuit while maintaining full functionality. The final circuit diagram is shown in figure 3.

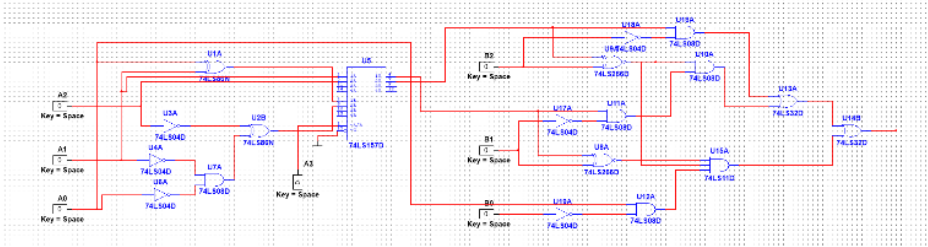


Fig. 3. The final circuit diagram (Photo/Picture credit: Original)

3 Critical path

In combinational digital circuit design, the critical path is the longest logical path that a signal must take from the input to the output without encountering a memory element such as a trigger. This path determines the maximum delay of the circuit and directly affects its operating frequency [11]. Optimizing critical paths is critical to improving the overall speed and efficiency of a logical system.

In the case of the 4-bit absolute value detector presented in this project, the critical path is determined by analyzing the gate level schematics. As shown in the figure 4, the critical path starts from non-gate U1A, passes through a series of logic gates such as U2A (AND), U3A (XOR), AND U4A (AND), then passes through U7A (OR), U8A (AND), U9A (AND), U10A (OR), and finally reaches U11A (OR). The path consists of multiple logic layers connected sequentially, forming the longest uninterrupted logic operation chain in the circuit.

This path is considered a critical path for several reasons. First, it contains the most logical levels, which results in the largest cumulative propagation delay. A delay is introduced as the signal passes through each gate, and the sequential nature of the path means that each delay is added directly to the total. Other paths in the circuit contain fewer gates or bypass certain operations, resulting in shorter delays and therefore do not limit the overall timing of the circuit.

Second, the path includes gates with relatively high inherent delays, such as XOR gates U3A, which typically have longer propagation times compared to basic and or gates due to their more complex internal structure. In addition, the chain includes multiple with gates (U2A, U4A, U8A, U9A) in series, which further contributes to delayed accumulation, especially if the gates are not optimally sized or if they drive significant capacitive loads.

In addition, this path is functionally critical as it is responsible for handling critical signals used in absolute value logic. It not only handles bitwise operations, but also

participates in determining the final output conditions. The logical flow from U1A to U11A performs a meaningful data transformation, and no output can be generated until all gates in the sequence have produced results.

To sum up, the path from U1A to U11A is the true critical path of the circuit, both in terms of gate depth and signal dependence. It finally sets a limit on the maximum clock speed of the system. In the next section, it will explore techniques to reduce this path delay, thereby improving the overall performance and responsiveness of the absolute value detector.

And the critical path is shown in Figure 4.

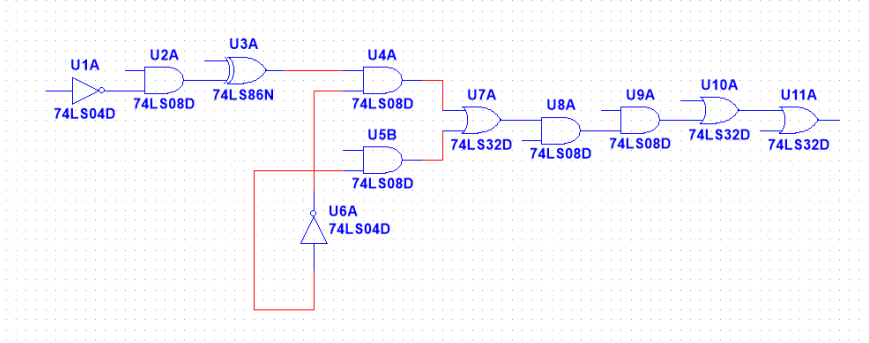


Fig. 4. The critical path (Photo/Picture credit: Original)

4 Optimization

4.1 Delay Analysis

Upon completing the design, an in-depth examination of the 4-bit absolute value detector's timing characteristics was performed. In digital circuit design, delay serves as a key indicator of system performance, as it governs the overall processing speed. To evaluate this, each logic gate within the circuit was modelled, enabling the calculation of the total propagation delay from input to output. The path with the longest signal traversal time—referred to as the critical path—was determined based on the circuit's layout and structure.

Using the following formula to calculate the total delay:

$$\text{Delay} = \sum_{i=1}^N (p_i + g_i * h_i) \quad (3)$$

Stage effort:

$$f_i = g_i * h_i \quad (4)$$

Path electrical effort:

$$H_{\text{path}} = C_{\text{out}} \div C_{\text{in}} \quad (5)$$

Path logical effort:

$$G_{\text{path}} = \prod_{i=1}^N g_i \quad (6)$$

Use the formulas below to calculate the size of each gate in the critical path:

$$G = g_{inv}^1 * g_{2NAND}^3 * g_{XOR}^1 * g_{2NOR}^3 = 18.37, H = 32, B = 1 \quad (7)$$

$$F = GBH = 587.84, f = F^{\frac{1}{N}} = 2.07 \quad (8)$$

$$C_{in} = g * \frac{C_{out}}{f^*} \quad (9)$$

Power Consumption Analysis

Energy efficiency is a key concern in contemporary digital circuit development. As integrated circuit technologies continue to evolve, the emphasis on low-power design has significantly increased. Total power consumption typically includes both static and dynamic components, with dynamic power being the predominant factor in most practical scenarios.

Assuming a threshold voltage $V_{th}=0.2V$, the following expressions are used to estimate power consumption:

$$t_p = \frac{CV_{DD}}{K'(V_{DD} - V_{TH})^2}, K' = \frac{W}{L} K \quad (10)$$

$$C = 32 * (C_{gn} + C_{gp}) = 34.56 fF, K' = 860 \mu A / V^2 \quad (11)$$

Since $\frac{t_p}{t_{p \min}} = 1.5$, assume $x = V_{DD}$, we can conclude that:

$$1.5 = \frac{\frac{x}{(x-0.6)^2}}{\frac{x_{\min}}{(x_{\min}-0.6)^2}} \quad (12)$$

$$\text{So, } V_{DD}^{opt} = 0.73v, D_1 = 1.5D = 57.84ps \quad (13)$$

Energy consumption when $V_{DD} = 1v$:

$$E = C_L V_{DD}^2 = 32 \quad (14)$$

The overall energy consumption of the circuit:

$$E = C_L V_{DD}^{opt^2} = 17.05 \quad (15)$$

5 Conclusion

The implementation of the 4-bit absolute value detector demonstrates a balanced integration of speed, accuracy and energy efficiency. Through the combination of static CMOS and through-tube logic (PTL), the circuit achieves the exact conversion of the signed binary input to its absolute value, and is verified against a comprehensive 16-case truth table. Logic optimization using Karnaugh diagrams and gate level integration significantly reduces circuit complexity. The critical paths identified from U1A to U11A are analysed using a logical effort approach to guide appropriate gate size and layout improvements. This helps minimize the overall propagation delay and increases

the speed of the circuit. Through hybrid logic design and signal multiplexing, power consumption is effectively controlled. While dynamic power is still a major factor, overall energy use is reduced by minimizing redundant logic and optimizing switching activity. The final circuit structure not only integrates the absolute value calculation, but also includes a comparator in the same logic flow, improving performance without increasing area costs. Overall, the project succeeded in delivering a compact and optimized design for low-power and high-speed digital systems that meet the growing needs of modern VLSI applications.

Despite the encouraging results of the study, some challenges remain. The current design and analysis are mainly based on theoretical modelling and logical effort estimation. While this approach provides valuable insights into latency and power trends, the lack of circuit-level simulation or physical implementation limits verification of actual performance. Without simulation tools such as SPICE, the variability of process parameters, parasitic capacitance, and temperature effects cannot be fully captured. In addition, the design assumes ideal switching behaviour and static power conditions, which may vary due to manufacturing constraints. Future work should therefore include detailed transistor-level simulations and layout-based parasitic extraction to assess the robustness and practical feasibility of the circuit. In addition, the project mainly focuses on logic-level optimization and does not explore emerging low-power technologies or new materials. With growing interest in carbon nanotube transistors, FINFETs, and 2D materials, future iterations could integrate these advanced technologies to further reduce power consumption and area. Perhaps as new technologies develop in the future, they can be explored in combination with hybrid logic styles, potentially leading to greater performance improvements and enabling such detectors to be deployed in ultra-low power and high-speed embedded systems.

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